

- » Understanding the specifics of bonds
- » Examining securities issued by the U.S. government
- » Looking at short-term bonds
- » Practicing with some sample questions

Chapter 7

Debt Securities: Corporate and U.S. Government Loans

Instead of giving up a portion of their company (via stock sales), corporations can borrow money from investors by selling bonds. Local governments (through municipal bonds; see Chapter 8) and the U.S. government also issue bonds. For SIE exam purposes, most bonds from the same issuer are considered to be safer than stocks.

Bondholders aren't owners of a company, as stockholders are; they're creditors. Bondholders who purchased bonds in the primary market are lenders of money to an institution for a fixed period and receive interest for doing so. This arrangement allows the institution to borrow money on its terms (with its chosen maturity date, scheduled interest payments, interest rate, and so on), which it can't do when borrowing from a lending institution.

The SIE exam tests you on your ability to understand the types of bonds issued, the terminology used, and (yes) some math. This chapter has you covered on topics relating to corporate and U.S. government debt securities. At the end of this chapter, I've added a quick chapter quiz to help you get a feel for questions you might see on the real SIE exam.

Tackling Bond Terms, Types, and Traits

Before you delve deeper into bonds, make sure that you have a good handle on the basics. Understanding bond basics is a building block that can make all the rest of the bond stuff easier. In this section, I first review basic bond terminology and then move on to some bond characteristics.

Covering bond terminology

The SIE exam designers expect you to know general bond terminology. (And I give it to you here; that's why I get paid the big bucks!) If you've already studied another textbook or taken a course, this section serves as a review. This stuff is basic, but the SIE exam does test it:



REMEMBER

» **Maturity date:** All issued bonds have a stated maturity (20 years, 30 years, and so on). The *maturity date* is the date bondholders get paid back for the loans they made. At maturity, bondholders receive par value (see the next item) plus any interest due. Because not as many investors are looking to tie up their money for a long period, short-term bonds are typically more liquid (actively traded) than long-term bonds.

» **Par value:** *Par value* is the face value or denomination of the bond. Although par value isn't significant to common stockholders (whose issuers use it solely for bookkeeping purposes), it's important to bondholders. For SIE exam purposes, you can assume that the par value for each bond is \$1,000 unless otherwise stated in the question.

Bond prices are quoted as a percentage of par value, most often without the percent sign. A bond trading at 100 is trading at 100 percent of \$1,000 par. Regardless of whether investors purchase a bond for \$850 (85), \$1,000 (100), or \$1,050 (105), they'll receive par value plus any interest due at the maturity date of the bond, usually with semiannual interest payments along the way. Corporate bonds are usually quoted in increments of $\frac{1}{8}$ percent ($\frac{1}{8}\% = 0.00125$ or \$1.25), so a corporate bond quoted at 99 $\frac{3}{4}$ % (99.375 percent) would be trading at \$993.75.

If you see something like "the last transaction in ABC 7.50s 2035 was at 101", this means that ABC bonds with a coupon rate of 7.50 and an expiration of 2035 last traded at \$1,010 (101% of 1,000 par). This means that the bonds traded at a premium (over par value).

» **Coupon rate:** Investors aren't lending money to issuers for nothing, of course; they receive interest for providing loans to the issuer. The *coupon rate* on the bond tells the investors how much annual interest they'll receive. Although bonds are no longer issued with physical coupons, some bonds previously required investors to detach dated coupons (bearer bonds and partially registered bonds) from their bonds and turn them in to receive their interest payments.

The coupon rate is expressed as a percentage of par value. A bond with a coupon rate of 6 percent, for example, would pay annual interest of \$60 ($6\% \times \$1,000$ par value). You can assume that bonds pay interest semiannually (twice a year) unless otherwise stated. So, in this example, an investor would receive \$30 every six months for each bond owned.



REMEMBER

Bondholders receive *interest* (payment for the use of the money loaned), and stockholders receive *dividends*. (See Chapter 6.)

» **The bond indenture:** The *indenture* (also known as *deed of trust or resolution*) is the legal agreement between the issuer and its bondholders and is printed on or attached to the bond certificate. All indentures contain basic terms:

- The maturity date
- The par value
- The coupon rate (interest rate) and interest payment dates
- Any collateral securing the bond (See "Comparing secured and unsecured bonds" later in this chapter.)
- Any callable or convertible features (Check out "Contrasting callable and put bonds" and "Popping the top on convertible bonds" later in this chapter.)

The bond indenture also includes the name of a trustee. A *trustee* is an organization that administers a bond issue for an institution; it ensures that the bond issuer meets all the terms and conditions associated with the borrowing. Essentially, the trustee tries to make sure that the issuer does the right thing.

The following question tests your knowledge of bond interest.



EXAMPLE

Melissa R. purchased 100 AA-rated bonds issued by COW Corp. Melissa purchased the bonds at 105 percent of par value, and they're currently trading in the market at 104. If the coupon rate is $7\frac{1}{2}$ percent, how much annual interest does Melissa receive?

- (A) \$37.50
- (B) \$75.00
- (C) \$3,750.00
- (D) \$7,500.00

The correct answer is (D). This question is a nice, easy one after you wade through the information you don't need. You need only the number of bonds and the coupon rate to figure out the answer. Don't let yourself get distracted by the AA rating, the purchase price, or the market price; that information is there to confuse you.

Melissa purchased 100 bonds at \$1,000 par (remember that you can assume \$1,000 par) with a coupon rate of $7\frac{1}{2}$ percent, so do the math:

$$100 \text{ bonds} \times \$1,000 \text{ par} \times 7\frac{1}{2} / 2 = \$7,500.00$$

Option (C) would have been correct if the question had asked for the semiannual interest.

Following bond issue and maturity schedules

Bond certificates can not only be in different forms, but also be scheduled with different types of maturities. Maturity schedules depend on the issuer's needs. The following list presents an explanation of the types of bond issues and maturity schedules:

» **Term bonds:** Term bonds are all issued at the same time and have the same maturity date. Short term corporate bonds have initial maturities of 3 years or less; medium term, 4-10 years; long term more than 10 years. If a company issues \$20 million worth of term bonds, for example, they all mature in 30 years. Because of the large payment that's due at maturity, most corporations that issue this type of bond have what is called a *sinking fund*. (More on that in the next paragraph.) Most corporations issue term bonds because they're able to lock in a set coupon rate for a set period of time.

A corporation creates a sinking fund when it sets aside money over time in order to retire its debt. Investors like to see that a sinking fund is in place because it lowers the likelihood of *default* (the risk that the issuer can't pay interest or par value back at maturity). The likelihood of default is determined by credit rating services such as Moody's and Standard & Poor's; for more on that topic, see "Considering bond credit ratings" later in this chapter.

» **Series bonds:** These bonds are issued in successive years but have only one maturity date. Issuers of series bonds pay interest only on the bonds that they've issued so far. Construction companies that are building developments in several phases are most likely to issue this type of bond. Fewer series bonds are issued than term and serial bonds.

» **Serial bonds:** In this type of bond issue, a portion of the outstanding bonds matures at regular intervals. (Perhaps 10 percent of the entire issue matures yearly.) Serial bonds are usually issued by corporations and municipalities to fund projects that provide regular income streams. Most municipal (local government) bonds are issued with serial maturities.

A serial bond that has more bonds maturing on the final maturity date is called a *balloon issue*.



TIP

The SIE exam focuses mainly on term and serial bonds. A typical SIE exam question may ask “Which of the following types of bonds is most likely to have a sinking fund?” The answer is term bonds; they’re the ones in which the entire bond issue has one maturity date and, therefore, require the issuer to pay a lot of money at one time.

Comparing secured and unsecured bonds

The assets of the issuer may or may not back bonds. For test purposes, assume that bonds backed by *collateral* (assets that the issuer owns) are considered to be safer for the investor. Such *secured bonds* (bonds backed by collateral) involve a pledge from the issuer that a specific asset, such as property, would be sold to pay off the outstanding debt in the event of default. *Unsecured bonds*, as the name implies, involves no such pledge. Obviously, with all else being equal, secured bonds normally have a lower yield than unsecured bonds.

The SIE exam tests your knowledge of several types of secured bonds:

- » **Mortgage bonds:** These bonds are backed by property that the issuer owns. In the event of default or bankruptcy, the issuer must liquidate the property to pay off the outstanding bonds. Mortgage bonds may be open- or closed-end. With an *open-end* mortgage bond, the issuer may borrow more money by using the same property as collateral. With a *closed-end* mortgage bond, the issuer can’t borrow more money by using the same property as collateral.
- » **Equipment trusts:** These bonds are issued mainly by transportation companies and are backed by equipment they own (airplanes, oil rigs, trucks, and so on). If the company defaults on its bonds, a trustee would sell the assets backing the bonds to satisfy the debt.
- » **Collateral trusts:** These bonds are backed by financial assets (stocks and bonds) that the issuer owns. A *trustee* (a financial institution the issuer hires) holds the assets and would sell them to pay off the bonds in the event of default.
- » **Guaranteed bonds:** Guaranteed bonds are backed by a firm other than the original issuer, often a parent company. If the issuer defaults, the guarantor pays off the bonds. As such, the rating of the bonds is tied to the rating of the guarantor.

As I imply previously, unsecured bonds are the opposite of secured bonds: They aren’t backed by any assets whatsoever — only by the good faith and credit of the issuer. If a reputable company that has been around for a long time issues the bonds, the bonds aren’t considered to be too risky. If they’re issued by a relatively new company or one with a bad credit rating, hold onto your seat! Again, for SIE exam purposes, assume that unsecured bonds are riskier than secured bonds. Here’s the lineup of unsecured bonds:

- » **Debentures:** These bonds are backed only by the issuer’s good word and written agreement (the indenture) stating that the issuer will pay the investor interest when due (usually, semiannually) and reach par value at maturity.
- » **Income (adjustment) bonds:** These bonds are the riskiest of all. The issuer promises to pay par value back at maturity and will make interest payments only if earnings are high enough. Companies in the process of reorganization usually issue these bonds at a deep discount. (The bonds might sell for \$500 and mature at par, or \$1,000, for example.) For test and real-world purposes, you shouldn’t recommend these bonds to investors who can’t afford to take a lot of risk.



Because secured bonds are considered to be safer than unsecured bonds, secured bonds normally have lower coupon rates. You can assume that for the SIE, the more risk an investor takes, the more reward they will receive. Remember the saying “More risk equals more reward.” More reward may be in the form of a higher coupon rate or a lower purchase price. Either or both may lead to a higher yield for the investor.

Check out the following question for an example of how the SIE may test your knowledge of the types of bonds.



Jon Bearishnikoff is a 62-year-old investor who has 50 percent of their portfolio invested in common stock of up-and-coming companies. The other 50 percent of their portfolio is invested in a variety of stocks of more-secure companies. Jon would like to start investing in bonds. If Jon’s main concern is the safety of the investment, which of the following bonds would you least likely recommend?

- (A) Collateral trust bonds
- (B) Mortgage bonds
- (C) Equipment trust bonds
- (D) Income bonds

The answer you’re looking for is (D). This problem includes a lot of garbage information that you don’t need to answer the question. One of your jobs (should you decide to accept it) is to dance your way through the question and cherry-pick the information that you do need. The last sentence is usually the most important one for answering a question. Jon is looking for safety; therefore, you’d be least likely to recommend income bonds because they’re usually issued by companies in the process of reorganizing. As a side note, if you become Jon’s broker, ensure they don’t have 100 percent of their investments in stock. At age 62, Jon should have a decent amount of their portfolio invested in fixed-income securities such as bonds.



When you’re comparing short-term and long-term debt securities, short-term bonds from the same issuer are considered to be safer because the investor isn’t tying up their money for as long a period. Because of the extra risk long-term bondholders are taking for tying up their money for a longer period, long-term bondholders generally (except in rare cases) receive a higher coupon (interest rate) for taking that additional risk.

Making Basic Bond Price and Yield Calculations

Although there’s not a lot of math on the SIE exam, it does test your knowledge of bond prices and bond yields and how to calculate them. In this section, I review the relationship between bond prices and bond yields. Outstanding bond prices typically don’t remain static. As you can imagine, they, too, are affected by things like supply and demand, corporate rating change (see “Determining the Best Investment: Comparing Corporate Bonds” later in this chapter), interest rate changes, whether the bond was purchased at a discount (below par value) or a premium (above par value), and so on.



The relationship between outstanding bond prices and yields is an *inverse* (opposite) one. You can assume for SIE exam purposes that if interest rates decrease, outstanding bond prices increase, and vice versa. Suppose that a company issues bonds with a 4 percent coupon rate for \$1,000. After the bonds are on the market, interest rates decrease. Now the company can issue bonds with

a 3.5 percent coupon rate. Investors with the 4 percent bonds are in a very good position and can demand a premium for their bonds should they decide to sell them in the market.

Rates down ↓↓ = Prices up ↑↑

Rates up ↑↑ = Prices down ↓↓

As a side note, when an investor sells their bonds between coupon dates, they're entitled to accrued interest. Suppose that this investor owned a bond paying \$20 interest every 6 months. If they sold that bond halfway between coupon dates, they'd be entitled to receive \$10 from the purchaser on top of the selling price. Calculations for accrued interest (broken down on a daily basis) aren't part of the SIE exam but are covered in detail in some of the top-off exams, such as the Series 7.

The following sections review the types of bond yields and how the SIE exam tests this topic.

Nominal yield (coupon rate)

The *nominal yield* (NY) is the easiest yield to understand because it's the coupon rate on the face of the bonds. For exam purposes, you can assume that the coupon rate will remain fixed for the life of a bond. If you have a 5 percent bond, the bond will pay \$50 per year interest ($5\% \times \$1,000$ par value) for the life of the bond. When a problem states that a security is a 5 percent (or 6 percent or whatever) bond, it's giving the nominal yield.

Current yield

The *current yield* (CY) is the annual rate of return on a security. The CY of a bond changes when the market price changes. You can determine the CY by dividing the annual interest by the market price:

$$\text{Current yield (CY)} = \frac{\text{annual interest}}{\text{market price}}$$

The following question involves bond yields.



EXAMPLE

Monique Moneybags purchased one XYZ convertible mortgage bond at 105. Two years later, the bond is trading at 98. If the coupon rate of the bond is 6 percent, what is the current yield of the bond?

- (A) 5.7%
- (B) 6.0%
- (C) 6.1%
- (D) Cannot be determined

The correct answer is (C). Yes, I'm giving you a question with a lot of unnecessary information. All I can tell you is that unfortunately, you'll have to get used to it. The securities exam creators are notorious for inserting useless (and sometimes misleading) information into the questions to daze and confuse you. In this case, you need only the annual interest and the market price to calculate the answer. Use the following formula to get your answer:

$$\text{CY} = \frac{\text{annual interest}}{\text{market price}} = \frac{\$60}{\$980} = 6.1\%$$

The annual interest is \$60 (6% coupon rate $\times$ \$1,000 par value), and the current market price is \$980 (98% of \$1,000 par). The facts that the bond is convertible (bondholders can trade it for common stock; see "Popping the top on convertible bonds" later in this chapter) or a mortgage bond (backed by the issuer's property) and that it was purchased at 105 (\$1,050) are irrelevant.



REMEMBER

“Cannot be determined,” as tempting as it may be, is almost never the correct answer on the SIE exam.

Yield to maturity (basis)

The *yield to maturity* (YTM) is the yield an investor can expect if holding the bond until maturity. The YTM takes into account not only the market price, but also the par value, the coupon rate, and the amount of time until maturity. When someone yells to you, “Hey, what’s that bond yielding?” (all right, maybe I run in a different circle of friends), they’re asking for the YTM. The formula for YTM is as follows:

$$\text{YTM} = \frac{\text{annual interest} + \text{annual accretion or} - \text{annual amortization}}{(\text{market price} + \text{par value}) / 2}$$

$$\text{annual accretion} = \frac{\text{par value} - \text{market price}}{\text{years until maturity}}$$

$$\text{annual amortization} = \frac{\text{market price} - \text{par value}}{\text{years until maturity}}$$

I wouldn’t worry about committing the preceding equations to memory. The amount of math you’ll need on the SIE exam is pretty small. But it’s important that you understand the concepts of measurement and the meanings of terms such as YTM, yield to call, total return, and yield to worst.



TIP

Accretion and amortization (the adjustment of the bond price toward par over the amount of time until the bond matures), as in the earlier equation, aren’t tested on the SIE exam, but you need to be able to calculate them on some of the corequisite exams. (Accretion is used when a bond is purchased at a price below par value and amortization is used when a bond is purchased at a price above par value.) The preceding equation should help you visualize how YTM works.

Yield to call

The *yield to call* (YTC) is the amount that the investor receives if the bond is called by the issuer prior to maturity, which requires the bondholders to surrender their bonds for par value and sometimes above. The calculations are similar to those for the YTM (see the preceding section), but you substitute the call price for the par value. Your chances of needing to know this on the SIE exam are even more remote than those of needing the YTM calculations.

Yield to worst

To determine the *yield to worst* (YTW), you have to calculate the yield to maturity and YTC for all the call dates (if there’s more than one) and choose the lowest. If you get a question on YTW, knowing the definition should be enough to get you by.

Total return

The *total return* calculates the full return on a particular investment over a given period of time. As with the other yields noted, you’ll likely not need to calculate it, but you’ll need to understand it. The total return provides you a percentage of gain or loss of an investment. To determine the total return, you need to follow these steps:

1. Determine the initial cost of the investment.
2. Calculate the total amount of interest or dividends received over the time of investment.

3. Add the interest or dividends to the selling price.
4. Divide that number by the initial cost of the investment and subtract 1.

Basis point

A *basis point* is one of the easier calculations. Basis points are typically used in the bond market, mutual funds, and exchange-traded funds (ETFs). Basis points are simply $\frac{1}{100}$ th of a percent. If the yield on a Treasury bond (T-bond) lowers 1.2 percent to 1.1 percent, it is said to have moved ten basis points. Also, when comparing expenses on a fund (see Chapter 9), people often refer to the fund expenses by basis points. If you're comparing two funds, one fund's expenses might be 0.15 percent and the other's 0.13 percent for a difference of 0.02 percent or two basis points.

Accrued interest

When investors purchase outstanding bonds in the market, they may have to pay an additional cost besides the market price (and, of course, your commission). The additional cost is called accrued interest. *Accrued interest*, which is due when bonds are purchased between coupon dates, is the portion of the interest still due to the seller. As you may remember, most bonds pay interest once every six months. If an investor holds onto a bond for 87 days out of a 180-day six-month period (for corporate and municipal bonds), they are entitled to $87/180$ of that next interest payment; that's accrued interest.



REMEMBER

Although you won't be required to calculate accrued interest when taking the SIE (not so with the Series 7), you should understand the concept and that for *corporate* and *municipal bonds*, accrued interest is calculated using *30-day months* and *360 day years*. When calculating accrued interest on *U.S. government bonds and notes*, you have to use *actual days in a month* and a *365-day year*.

Accrued interest is calculated from the trade date, up to, but not including the settlement date.

Determining the Best Investment: Comparing Corporate Bonds

As you grind your way through SIE exam questions, you may be asked to determine the best investment for a particular investor. You need to look at the question carefully for clues to help you choose the correct answer. (Is the investor looking for safety, for example, or is the investor close to retirement?) Consider several factors, including credit rating, callable and put features, and convertible features. Certainly, the investor's investment objectives (if stated) have to take center stage.

Considering bond credit ratings

The institutions that rate bonds are most interested in the likelihood of *default* (the likelihood that the interest and principal won't be paid when due). For the exam, you can assume that the higher the credit rating, the safer the bond and, therefore, the lower the yield.

The two main bond credit rating companies are Moody's and Standard & Poor's (S&P). S&P ratings of BB and lower and Moody's ratings of Ba and lower are considered to be *junk bonds* or *high-yield bonds*, which have a higher likelihood of default, as Table 7-1 explains. (Another credit rating

service, called Fitch, uses the same rating symbols as S&P.) *Note:* Different sources may show some slight variations in how S&P and Moody's ratings compare, but the relationships shown here are the most common.

TABLE 7-1 Bond Credit Ratings (by Quality)

Quality	S&P	Moody's
Highest	AAA	Aaa
High	AA	Aa
Upper medium	A	A
Lower medium	BBB	Baa
Speculative (junk)	BB	Ba
Speculative (junk): Interest or principal payments missed	B	B
Speculative (junk): No interest being paid	C	Caa
In default	D	D

As if these categories weren't enough, S&P can break down each category even further by adding a plus (+) or minus (–) sign after the letter category. The plus sign represents the high end of the category, and the minus sign designates the lower end of the category. If you see no plus or minus sign, the bond is in the middle of the category. Moody's can break down a category further by adding a 1, 2, or 3. The number 1 is the highest ranking, 2 represents the middle, and 3 is the lowest. The top four ratings are considered to be *investment grade* (AAA, AA, A, and BBB for S&P; Aaa, Aa, A, and Baa for Moody's), and the letter ratings below that are considered to be *junk bonds* or *high-yield bonds*.



TIP

S&P uses all capital letters (AAA, AA, and so on). Additionally, S&P has an ampersand (&) between the "S" and the "P" in its name. Think of the ampersand as being like a plus sign to help you remember that S&P uses pluses and minuses within its categories.

Here's a typical bond-ratings question.



EXAMPLE

Place the following S&P bond ratings in order from highest to lowest:

- I. A+
 - II. AA
 - III. A–
 - IV. BBB+
- (A) I, II, III, IV
 (B) I, III, II, IV
 (C) IV, I, II, III
 (D) II, I, III, IV

The correct answer is (D). When you're answering this type of question, always look at the letters first. The only time pluses or minuses come into play is when two answers have the same letters, as in statements I and III. The highest choice is AA, followed by A+ because it's higher than A–, which is even higher than BBB+.

Contrasting callable and puttable (put) bonds

Your mission for the SIE exam is to know which bonds are better for investors and when bonds are likely to be called or put. As you may know, bonds can be issued in callable and put forms:

» **Callable bonds:** A *callable bond* is a bond that the issuer has the right to buy back (call from investor) at the price stated on the indenture (deed of trust). All callable bonds must have a *call provision* that stipulates the call date (the first date the bonds can be called), and the *call price* (the price investors will receive if the bonds are called). Callable bonds are riskier for investors because investors can't control how long they can hold onto the bonds. To compensate for this risk, they usually receive a higher coupon rate (more risk = more reward).

Most callable bonds are issued with call protection. *Call protection* is the amount of time (usually, several years) that an issuer has to wait before calling its bonds (such as five years after issuance). Some callable bonds also have a *call premium*, which is an amount over par value that an issuer has to pay if it calls its bonds in the year(s) immediately following the expiration of the call protection.

If a *make whole call provision* is included, it allows the issuer to call the bonds provided that the issuer makes a lump-sum payment to investors that includes not only payment for the bond but also the present value of any future interest payments investors will miss because of the call.

Another type of bond that can be callable is a step coupon bond. Also known as *stepped coupon bonds* or *step-up coupon securities*, *step coupon bonds* typically start at a low coupon rate, but the coupon rate increases at predetermined intervals, such as every five years. The issuer typically has the right to call the bonds at par value at the time the coupon rate is due to increase.

» **Put bonds:** *Put bonds* are better for investors. Put bonds allow the investor to "put" the bonds back (redeem them) to the issuer at any time at the price stated on the indenture. Because the investors have the control, put bonds are (of course) rarely issued. Because these bonds provide more flexibility to investors (who have an interest in the bond and stock prices), put bonds usually have a lower coupon rate.



REMEMBER

There's a direct correlation between interest rates and when bonds are called or put. Issuers call bonds when interest rates decrease; investors put bonds when interest rates increase. Check out the following question to see how this process works.



EXAMPLE

Issuers would more likely call their bonds when interest rates are

- (A) increasing
- (B) decreasing
- (C) staying the same
- (D) fluctuating

The correct answer is (B). Being adaptable when taking the SIE exam can certainly help your cause. In this question, you have to look from the issuer's point of view, not the investor's. An issuer would call bonds when interest rates decrease because it could redeem the bonds with the higher coupon payments and issue bonds with lower coupon payments to save money. Conversely, investors would put their bonds back to the issuer when interest rates increase so they could invest their money at a higher interest rate.



REMEMBER

You can assume for SIE exam purposes that if interest rates increase, bond yields increase.

Popping the top on convertible bonds

Bonds that are convertible to common stock are called *convertible bonds*. Convertible bonds are attractive to investors because investors have an interest in the bond price as well as the price of the underlying stock. *Parity* occurs when a convertible bond and its underlying stock (the stock it's convertible into) are trading equally (that is, when a bond trading for \$1,100 is convertible into \$1,100 worth of stock).



TIP

If a convertible bond is trading below parity (below the value of the underlying shares it covers), it would make financial sense for an investor to convert. It costs the investor nothing to convert their bonds to the underlying stock.

In the event that you have to answer a question about convertible bonds to determine whether they're worth converting to the underlying stock, you can use the following formula:

$$\text{conversion ratio} = \frac{\text{par value}}{\text{conversion price}}$$



REMEMBER

You see this same equation in Chapter 6 relating to convertible preferred stock.

You can use the conversion ratio to calculate the parity price:

$$\text{parity price of the bond} = \text{market price of the stock} \times \text{conversion ratio}$$

or

$$\text{parity price of the stock} = \frac{\text{market price of the bond}}{\text{conversion ratio}}$$



EXAMPLE

ABCD convertible bonds are convertible into 25 shares of common stock. If the stock trades at 36, what is the parity price of the bond?

- (A) \$36
- (B) \$40
- (C) \$80
- (D) \$900

The correct answer is (D). When you get a parity-price question, you have to determine where the stock and the bond (or convertible preferred stock) are trading equally. In this case, you were already given the conversion ratio, which is the amount of shares the bond is convertible into (25). So using that information, you can plug the information given into the following equation:

$$\text{parity price of the bond} = \text{market price of the stock} \times \text{conversion ratio}$$

$$\text{parity price of the bond} = \$36 \times 25 \text{ shares} = \$900$$

Exploring U.S. Government Securities

On the SIE and companion exams, you need to know the basic types of U.S. government securities, their initial maturities, and certain characteristics.

As you may already know, the U.S. government also issues bonds. U.S. government bonds are considered to be the safest of all securities. (Yes, you did read that correctly: the *safest of all securities*. I feel that this statement is worth repeating.) How can U.S. government securities be so safe when the government is running such a large deficit? Guess what — I don't know, and you don't need to know either. I can only assume that the U.S. government can always print more

currency to make payments on its securities if needed. But even U.S. government securities are subject to certain risks, such as interest risk, reinvestment risk, purchasing power risk, and so on. (See Chapter 13.)

These days, all U.S. government securities are issued and held in electronic (book-entry) form. Because Treasury bonds have maturities of up to 30 years, however, some are still out there in paper form.



TIP

With government bonds, you use some of the same types of calculations you use for corporate bonds.

Understanding the types of U.S. government securities

Table 7-2 gives you an overview of types of U.S. government securities and their specifics. Memorize all the information in the following table so that you can ace U.S. government securities questions on the SIE exam. Individual investors can purchase U.S. government securities directly through www.treasurydirect.gov, through a bank, or through a broker. The securities are issued in electronic form, so investors don't receive the actual bond certificate.

TABLE 7-2 U.S. Government Securities and Time until Maturity

Security	Initial Maturity	Characteristics
Treasury bills (T-bills)	4, 8, 13, 17, 26, or 52 weeks; considered to be short-term U.S. debt securities	Issued at a discount and mature at par. The difference between the purchase price and par is considered to be interest, even though no interest payments were made. Minimum purchase is \$100.
Treasury notes (T-notes)	2, 3, 5, 7, or 10 years; considered to be intermediate-term U.S. debt securities	Pay interest every 6 months. Minimum purchase is \$100.
Treasury bonds (T-bonds)	20 or 30 years; considered to be long-term U.S. debt securities	Pay interest every 6 months. Minimum purchase is \$100.
T-STRIPS (Separate Trading of Registered Interest and Principal of Securities)	6 months to 30 years	Considered to be zero-coupon securities (issued at a discount and mature at par) and don't receive interest payments. Purchase price varies. The minimum face value must be \$100.
TIPS* (Treasury Inflation-Protected Securities)	5, 10, or 30 years	Pay interest every 6 months. Par value and interest payments adjust according to inflation or deflation. Minimum purchase is \$100.

**TIPS are tied to the Consumer Price Index (CPI), which measures inflation. The par value changes according to inflation. If inflation is high (prices of goods and services are increasing), the par value increases. If we're in a period of deflation (prices on goods and services are decreasing), the par value decreases. Because investors are getting a percentage of par value as their interest payments, the interest payments vary along with the par value.*



REMEMBER

For the SIE exam, keep in mind that the interest received on U.S. government securities is exempt from state and local taxes. The interest received on municipal bonds is exempt from federal taxes (although I get into that topic a little more in Chapter 8). Chapter 15 gives you the scoop on taxes.

Looking at agency securities

Agency bonds are ones issued by a U.S. government-sponsored agency or government-sponsored entity (GSE). The bonds are backed by the U.S. government, but not all are guaranteed by the full faith and credit of the U.S. government except for Government National Mortgage Association (GNMA) bonds, which are directly backed. As such, agency bonds (although almost as safe) are considered to be riskier than U.S. government bonds and notes such as T-bonds, T-notes, and T-bills. Not all U.S. government agency securities are exempt from SEC registration. GSEs include the following:

- » **GNMA** (Government National Mortgage Association, or Ginnie Mae): GNMA is the only agency securities backed by the full faith and credit of the U.S. government. GNMA supports the U.S. Department of Housing and Urban Development (HUD). GNMA is considered *asset-backed securities* (backed by mortgages), which typically are sold based on their *average life expectancy*, rather than their stated life expectancy (typically 30 years). Even though many people have a 30-year mortgage, the average life might be only 17 years or so based on refinancing and people selling their homes.
- » **FNMA** (Federal National Mortgage Association, or Fannie Mae): FNMA is a publicly held corporation that is responsible for providing capital for certain mortgages. As such, FNMA may purchase conventional mortgages, Department of Veterans Affairs mortgages, U.S. Federal Housing Association mortgages, and so on. FNMA is privately owned and publicly held but is still a GSE.
- » **FHLMC** (Federal Home Loan Mortgage Corporation, or Freddie Mac): As a public corporation, Freddie Mac was designed to create a secondary market for mortgages. Freddie Mac purchases residential mortgages from financial institutions and packages them into mortgage-backed securities that are sold to investors.
- » **FCS** (Farm Credit System): The FCS consists of lending institutions that provide financing and credit to farmers. It's a GSE but is privately owned. The FCS sells securities to investors and in turn loans the funds raised to farmers. The FCS is overseen by the Farm Credit Administration.
- » **SLMA** (Student Loan Marketing Association or Sallie Mae): SLMA isn't involved in providing mortgages but provides a secondary market for student loans. As such, SLMA purchases student loans and repackages them as short- and medium-term debt securities for sale to investors.



TIP

Certain mortgage-backed securities are susceptible to reinvestment risk because many homeowners refinance when interest rates fall (*prepayment risk*). In that case, holders of mortgage-backed securities get paid back sooner than expected and are reinvesting at a lower interest rate. In addition, if interest rates stay the same or increase, homeowners won't refinance as often, and holders of mortgage-backed securities may end up having to hold their investment for a longer period than expected (*extension risk*).

Playing It Safe: Short-Term Loans or Money Market Instruments

Your SIE exam will likely include a question or two on money market instruments. *Money market instruments* are relatively safe short-term loans that can be issued by corporations, banks, the U.S. government, and municipalities. Most of these instruments have maturities of one year or less;

they're usually issued at a discount and mature at par value. The following list reviews some basic characteristics of money market instruments to help you earn an easy point or two on the SIE exam:



REMEMBER

» **Repurchase agreements:** Repurchase agreements (*repos*) are a contract between a buyer and a seller. The seller of the securities (usually, T-bills) agrees to buy them back at a previously determined price and time. Repos are short-term loans.

» **Federal funds:** Federal funds are loans between banks to help meet reserve requirements. Federal funds are usually overnight loans for which the rates change constantly depending on supply and demand.

Reserve requirements are the percentage of deposits that member banks must hold each night as a regulatory protection device. Banks that aren't able to meet their reserve requirements may borrow from other banks at the fed funds rate. For more info on the fed funds rate and other tools that the Federal Reserve Board uses to influence money supply, see Chapter 13.

» **Corporate commercial paper:** Commercial paper is unsecured short-term corporate debt. Commercial paper is issued at a discount and matures at par value. Commercial paper is issued with an initial maturity of 270 days or less and is exempt from SEC registration.

» **Brokered (negotiable) certificates of deposit (CDs):** Brokered CDs are low-risk investments that originate from a bank and are outsourced to broker-dealers to sell to investors. Unlike typical CDs, which are purchased directly from a bank, brokered CDs can be traded in the market. Negotiable CDs that require a minimum investment of \$100,000 are often called *jumbo CDs*.

» **Eurodollars:** *Eurodollars* are American dollars held by a foreign bank outside the United States. This situation is usually the result of payments made to overseas companies. Eurodollars are not to be confused with Eurodollar bonds (dollar-denominated bonds issued and held overseas).

» **Banker's acceptances (BAs):** A BA is a time-draft (short-term credit investment) created by a company whose payment is guaranteed by a bank. Companies use BAs for importing and exporting goods.

» **T-bills:** The U.S. government issues T-bills at a discount, and they have initial maturities of 4, 8, 13, 17, 26, or 52 weeks. T-bills are somewhat unique in that they're sold and quoted on a discount-yield basis (YTM). U.S. government securities — and especially T-bills — are considered to be the safest of all securities.

Here's what a question on money market instruments might look like.



EXAMPLE

SNK Surfboard Company wants to import boogie boards from an Italian manufacturer in Sicily. SNK would use which of the following money market instruments to finance the importing of the boogie boards?

- (A) T-bills
- (B) Collateral trust bonds
- (C) Repurchase agreements
- (D) Banker's acceptances

The correct answer is (D). You can eliminate (B) right away because collateral trust bonds aren't money market instruments; they're secured long-term bonds. A BA is like a postdated check that's used specifically for importing and exporting goods.



TIP

Word association can help you here. If you see *importing*, *exporting*, or *time-draft*, your answer is probably BA.

Testing Your Knowledge

Now that you've discovered what you need to know about corporate and U.S. debt securities as far as the SIE exam goes, it's time to try some additional questions (besides the ones within the chapter). Read carefully so that you don't make any careless mistakes.

Practice questions

1. A corporate bond indenture includes which of the following?
 - I. The coupon rate
 - II. The maturity date
 - III. Par value
 - IV. Any collateral securing the bond

(A) I and II
(B) I, II, and III
(C) I, II, and IV
(D) I, II, III, and IV
2. Declan K. has 100 DEF corporate bonds with a coupon rate of $4\frac{1}{2}$ percent. The bonds were purchased at 98 percent of \$1,000 par each. How much interest will Declan receive the next time they get paid?

(A) \$2,205
(B) \$2,250
(C) \$4,410
(D) \$4,500
3. Which of the following types of corporate bond issues is most likely to have a sinking fund?

(A) Term
(B) Series
(C) Serial
(D) None of the above
4. All of the following are types of secured bonds EXCEPT
 - (A) mortgage bonds
 - (B) collateral trusts
 - (C) income bonds
 - (D) guaranteed bonds
5. Which of the following is the formula that determines the current yield of a bond?
 - (A) Current yield = Semiannual interest divided by the market price
 - (B) Current yield = Semiannual interest divided by the par value
 - (C) Current yield = Annual interest divided by the market price
 - (D) Current yield = Annual interest divided by par value

6. Melissa R. Corporation has issued \$6 million worth of 30-year callable bonds with a par value of \$1,000, a coupon rate of $5\frac{1}{4}$ percent, and 7 years call protection. Melissa R. Corporation would least likely call the bonds when interest rates are generally
- (A) increasing
 - (B) decreasing
 - (C) remaining the same
 - (D) none of the above
7. Which of the following are possible maturities for a U.S. Treasury bill?
- I. 4 weeks
 - II. 8 weeks
 - III. 16 weeks
 - IV. 26 weeks
- (A) I and II
 - (B) I, II, and III
 - (C) I, II, and IV
 - (D) I, II, III, and IV
8. Which of the following U.S. government securities has interest payments that vary according to inflation or deflation?
- (A) T-bills
 - (B) T-STRIPS
 - (C) TIPS
 - (D) T-notes
9. Which of the following securities are money market instruments?
- I. Banker's acceptances
 - II. T-bills
 - III. Commercial paper
 - IV. Treasury notes
- (A) I and II
 - (B) I, II, and III
 - (C) II, III, and IV
 - (D) I, II, III, and IV
10. Which of the following are direct obligations of the U.S. government?
- I. T-bills
 - II. GNMA
 - III. FNMA
 - IV. T-STRIPS
- (A) I and IV
 - (B) I, II, and III
 - (C) II, III, and IV
 - (D) I, II, and IV

- 11.** The last transaction in HIJ 6.50s 2035 was at 99. These HIJ bonds sold
- (A) at par value
 - (B) at a discount
 - (C) at a premium
 - (D) cannot be determined
- 12.** For callable bonds, the call provision must stipulate which of the following factors?
- (A) The date of maturity
 - (B) The market value
 - (C) The call date and coupon rate
 - (D) The call price and call date
- 13.** When issued, which of the following debt securities would mature in one year or less?
- (A) SLMA bonds
 - (B) Money market securities
 - (C) U.S. government-issued Treasury notes
 - (D) Call options
- 14.** When computing dollar prices and accrued interest on municipal bonds, they are typically calculated using
- (A) a 30-day month and a 360-day year
 - (B) a 30-day month and a 365-day year
 - (C) actual days in a month and a 360-day year
 - (D) actual days in a month and a 365-day year
- 15.** When would the stability of a portfolio of debt securities be the greatest?
- (A) During a time of rising interest rates
 - (B) During a time of falling interest rates
 - (C) If the maturities of the debt securities are long-term
 - (D) If the maturities of the debt securities are short-term
- 16.** These securities are typically traded based on their average life instead of their stated maturity dates.
- (A) U.S. government bonds
 - (B) Asset-backed securities
 - (C) Corporate callable bonds
 - (D) Industrial development bonds

Answers and explanations

1. **D.** The indenture of a bond includes the date that the bond matures, the coupon rate, par value (typically, \$1,000), collateral securing the bond (if any), and any callable or convertible features.
2. **B.** The coupon rate is based on the par value of the bonds, not the purchase price or market value. Declan purchased \$100,000 par value of bonds with a coupon rate of $4\frac{1}{2}$ percent. This means that Declan will receive \$4,500 ($4\frac{1}{2}\% \times \$100,000$) in interest per year. But you can assume (unless you're told differently in the question) that bonds pay interest semiannually (every six months). So you need to divide the annual interest by 2 to get \$2,250 ($\$4,500/2$).
3. **A.** Although all the choices listed may have a sinking fund (a fund for which the issuer sets aside money to pay the bonds off at maturity), term bonds are the ones most likely to have one. Term bonds are issues in which all the bonds are issued at the same time and have the same maturity date.
4. **C.** This question is an *except* question, so you're looking for the one that's not secured. Income bonds are ones issued by a corporation that's in trouble and needs to reorganize. In this case, the issuing corporation would issue these bonds at a deep discount and not make interest payments unless earnings are high enough. Income bonds are considered to be extremely risky and aren't suitable for most investors.
5. **C.** The current yield of a security is the annual rate of return divided by the market price of the security. So if the market price changes, so does the current yield.
6. **A.** A lot of information thrown into this question is meant to confuse you. The question just wants you to know when an issuer would least likely call its bonds. The whole idea of callable bonds is that issuers want the right to call their bonds if interest rates decrease because then they can issue bonds with lower coupon rates. So they'd least likely call their bonds when interest rates are increasing.
7. **C.** T-bills have initial maturities of 4, 8, 13, 17, 26, and 52 weeks. T-bills are short-term U.S. government securities that are issued at a discount and mature at par value.
8. **C.** TIPS pay interest every six months, like T-notes and T-bonds, but have an interest payment that increases or decreases depending on inflation or deflation.
9. **B.** Money market instruments are debt securities that mature in one year or less. They include repurchase agreements, federal funds, commercial paper (some may mature in more than a year), brokered CDs, Eurodollars, BAs, and T-bills.
10. **D.** All the choices listed are directly backed by the U.S. government except for FNMA securities. FNMA is a publicly held corporation that provides capital for certain mortgages. It's privately owned but is still government-sponsored.
11. **B.** Remember par value for a bond is 100 (100 percent of \$1,000 par). In this case, the bonds sold at a purchase price of 99 (99 percent of \$1,000), so they sold at a discount from par value.

- 12. D.** Although the indenture of the bond will provide the par value, coupon rate, and date of maturity, the call provision would provide the call price (what the issuer would pay the holder if calling the bonds) and also the first date that the bonds can be called.
- 13. B.** Although call options typically mature in one year or less, they aren't debt securities, so you can count that one out. Out of the other choices, the only debt securities that mature in one year or less at the time of issuance are money market securities.
- 14. A.** Logically, you should be able to throw out answers "B" and "C" because there is no way to make them work. If you calculate using 30-day months, it would have to work out to a 360-day year, not 365. Likewise, if you're using actual days in a month, it would have to work out to a 365-day year. Unlike U.S. government securities, which use actual days in a month and a 365-day year, municipal bonds typically compute using 30-day months and 360-day years.
- 15. D.** Remember, when interest rates change, short-term debt securities typically change more quickly in price, but long-term debt securities change more in price. Therefore, an investor holding shorter-term debt securities would be holding securities that are more stable in price than someone holding a portfolio of long-term debt securities.
- 16. B.** Asset-backed securities, such as GNMMAs, are typically traded based on their average life. Asset-backed securities are ones that are backed by assets such as loans, mortgages, and accounts receivable. So, for arguments sake, say an individual takes out a 30-year mortgage; the average life of that mortgage may be 17 years due to that individual moving or refinancing the loan.

